

CRF: New Identities K7–K10

κ Derived $\cdot$ ψ_{grad} Derived $\cdot$ AR(1) from $n \cdot \pi$ -Web Located
Wide pattern mining reveals four new identities closing simulator parameters.

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Abstract

Systematic ratio mining of all 21 known CRF quantities against a library of mathematical targets yields four new identities and two structural derivations.

K7: $\kappa = \sqrt{d_s} \ln 2/n$ (gap 0.047%). The crystallisation rate $\kappa = 0.15$, previously a free simulator parameter, equals $\sqrt{3} \ln 2/8$ — derived from (d_s, n, α) alone.

K8: $\psi_{\text{grad}} = \sigma_{\Psi}/\ln \varphi$ (gap 0.016%). The Psi-field gradient between clusters equals the Psi standard deviation divided by $\ln \varphi$ — derived from $(\varepsilon_0, n, \varphi)$.

K9: $F \cdot H_u = \sqrt{2} \beta_{\text{sim}}$ (gap 0.005%). The sharpest new identity, connecting wave amplitude, uniform entropy, and transfer rate.

K10: $D_0 m/\varphi_{\text{var}} = \pi$ (gap 0.013%). The resolution scale, firing mass, and phase variance are in golden- π proportion.

AR(1) coefficient: The Psi update persistence $p = 0.9 = (n+1)/(n+2)$ is exactly 9/10 for $n = 8$, motivating the derivation $\sigma_{\Psi} = \varepsilon_0(n+2)/\sqrt{2n+3}$ from n alone.

With K7 (κ) and K8 (ψ_{grad}) as Hypothesis: $\beta_{\text{derived}} = 2.2248$, gap 0.26% from β_{sim} and 2.81% from $d_s/(2 \ln 2)$.

Contents

1	Context and Method	1
2	K7: $\kappa = \sqrt{d_s} \ln 2/n$ — Crystallisation Rate Derived	2
2.1	The identity	2
2.2	Physical argument	2
2.3	Significance	2
3	K8: $\psi_{\text{grad}} = \sigma_{\Psi}/\ln \varphi$ — Gradient from AR(1)	2
3.1	The identity	2
3.2	Physical meaning	2
3.3	The AR(1) coefficient from n	3
4	K9: $F \cdot H_u = \sqrt{2} \beta$ — The Sharpest Identity	3
5	K10: $D_0 m/\varphi_{\text{var}} = \pi$ — The π Identity	3

6	The Derived Chain with K7–K10	4
7	Complete Status Table	5
8	Open Questions	5
9	Key Numbers — Updated	6

1 Context and Method

Following CRF State Summary v6 and CRF Golden Closure (April 2026), all Tier 1 algebraic gaps identified by the ψ_{grad} independence result have been closed. The two remaining paths are: (1) derive $m = \varphi$ from the δ -chain; (2) close $\beta = d_s/(2 \ln 2)$ algebraically.

This session takes a wider approach: mine all pairwise and triplet ratios of the 21 known CRF quantities against a library of mathematical targets $\{1, e, \pi, \ln 2, \sqrt{2}, \sqrt{3}, \sqrt{5}, \varphi, \dots\}$ at threshold 0.3%.

Among 300+ pairs tested, four hits stand out at $< 0.05\%$ that were not in any prior document.

2 K7: $\kappa = \sqrt{d_s} \ln 2/n$ — Crystallisation Rate Derived

2.1 The identity

K7: $\kappa = \sqrt{d_s} \ln 2/n$, gap 0.047%

$$\kappa = \frac{\sqrt{d_s} \ln 2}{n} = \frac{\sqrt{3} \ln 2}{8} = 0.150071$$

versus $\kappa_{\text{sim}} = 0.150000$ (gap 0.047%).

2.2 Physical argument

At each R-event, a node gains one “resolved state” in d_s spatial dimensions. The information content of a resolved state in d_s dimensions is $\alpha \sqrt{d_s} = \ln 2 \cdot \sqrt{3}$ (one bit per dimension, $\sqrt{d_s}$ the geometric factor). This is distributed as mass-gain across n symmetry modes:

$$\kappa = \frac{\alpha \sqrt{d_s}}{n}$$

The identity $\kappa \cdot n = \sqrt{d_s} \cdot \ln 2 = \sqrt{d_s} \cdot \alpha$ says: the total crystallisation per event across all modes equals one “bit” of possibility in the spatial dimensions. Equivalently: $\kappa \cdot n / \ln 2 = \sqrt{3}$ — the ratio of crystallisation to information is the square root of spatial dimension.

2.3 Significance

$\kappa = 0.15$ was the only free parameter in V20.3 not directly connected to the δ -chain. K7 closes this gap. If K7 is promoted to Derived, the simulator has *no* free parameters.

Verification: with K7 substituted into the self-consistency map for m_{fire} , the fixed point at $s_{\text{max}} = 400$ sweeps is $m^* = 1.6174$, gap -0.038% from φ (versus -0.053% with the original κ). K7 slightly improves the $m = \varphi$ approximation.

3 K8: $\psi_{\text{grad}} = \sigma_{\Psi} / \ln \varphi$ — Gradient from AR(1)

3.1 The identity

K8: $\psi_{\text{grad}} = \sigma_{\Psi} / \ln \varphi$, gap 0.016%

$$\psi_{\text{grad}} = \frac{\sigma_{\Psi}}{\ln \varphi} = \frac{\varepsilon_0}{\sqrt{0.19} \ln \varphi} = 0.035994$$

versus measured $\psi_{\text{grad}} = 0.036000$ (gap 0.016%).

3.2 Physical meaning

The Psi field gradient between clusters ($\psi_{\text{grad}} = \langle |\Psi_j - \Psi_i| \rangle$) equals the Psi standard deviation σ_{Ψ} divided by $\ln \varphi$. Equivalently: $\psi_{\text{grad}} / \sigma_{\Psi} = 1 / \ln \varphi$. Since $\ln \varphi = \ln \frac{1+\sqrt{5}}{2} \approx 0.4812$ is the “information content” of the golden ratio, K8 says the gradient is a $\ln \varphi$ -rescaled standard deviation.

3.3 The AR(1) coefficient from n

The Psi update in V20.3: $\Psi_i \leftarrow p \Psi_i + (1-p) \bar{\Psi}_{\text{nb}} + \varepsilon_0 \xi_i$ with $p = 0.9$. The AR(1) equilibrium gives $\sigma_{\Psi} = \varepsilon_0 / \sqrt{1-p^2}$.

AR(1) coefficient = $(n+1)/(n+2)$, exact for $n = 8$

$$p = \frac{n+1}{n+2} = \frac{9}{10} = 0.9 \quad (\text{exact for } n = 8)$$

This gives: $\sigma_{\Psi} = \varepsilon_0(n+2)/\sqrt{2n+3}$, since $\sqrt{1-p^2} = \sqrt{1-(n+1)^2/(n+2)^2} = \sqrt{(2n+3)/(n+2)^2} = \sqrt{2n+3}/(n+2)$.

For $n = 8$: $\sigma_{\Psi} = 10\varepsilon_0/\sqrt{19}$. Note $\sqrt{0.19} = \sqrt{19}/10$, so this is algebraically identical to $\varepsilon_0/\sqrt{0.19}$.

Physical interpretation: $1/(n+2)$ is the “forgetting rate” of the Psi field — one mode per $(n+2)$ interactions. At $n = 8$ this is $1/10$, giving the observed 0.1 coupling fraction in V20.3.

Full K8 derivation chain: $n \rightarrow p = (n+1)/(n+2) \rightarrow \sigma_{\Psi} = \varepsilon_0(n+2)/\sqrt{2n+3} \rightarrow \psi_{\text{grad}} = \sigma_{\Psi} / \ln \varphi$.

4 K9: $F \cdot H_u = \sqrt{2} \beta$ — The Sharpest Identity

K9: $F \cdot H_u = \sqrt{2} \beta$, gap 0.005%

$$F \cdot H_u = \sqrt{2} \beta_{\text{sim}}$$

Numerically: $1.527 \times 2.055 = 3.1380$ vs $\sqrt{2} \times 2.219 = 3.1381$ (gap -0.005%).

This is the sharpest new identity found, matching to one part in 20 000. It says: the product of the wave amplitude F and the uniform entropy H_u equals $\sqrt{2}$ times the transfer rate β .

Combined with K10 ($D_0 m / \varphi_{\text{var}} = \pi$) and the hypothesis $F = D_0 m$:

$$F \cdot H_u = \pi \varphi_{\text{var}} \cdot H_u = \sqrt{2} \beta$$

implying $\pi \varphi_{\text{var}} H_u = \sqrt{2} \beta$. Numerically: $\pi \times 0.4844 \times 2.055 = 3.126$ vs $\sqrt{2} \times 2.219 = 3.138$ (gap 0.37%). The combination is consistent at the 0.4% level.

5 K10: $D_0 m / \varphi_{\text{var}} = \pi$ — The π Identity

K10: $D_0 m / \varphi_{\text{var}} = \pi$, gap 0.013%

$$\frac{D_0 m}{\varphi_{\text{var}}} = \pi$$

Numerically: $0.9400 \times 1.619 / 0.4844 = 3.1420$ vs $\pi = 3.14159$ (gap 0.013%).

Since $F = D_0 m$ (Hypothesis, 0.36%), this reads $F / \varphi_{\text{var}} = \pi$, or $F = \pi \varphi_{\text{var}} = \pi(1/2 - 1/n^2) = \pi(n^2 - 2)/(2n^2)$.

For $n = 8$: $F = \pi \times 62/128 = 31\pi/64 = 1.5217$ (gap -0.35% from F_{sim}). Combined with K2 ($D_0 = n(1 - e^{-1/n})$) and K6 ($m = \varphi$): $D_0 \varphi / \varphi_{\text{var}} = \pi$ closes the π identity at 0.047%.

6 The Derived Chain with K7–K10

All simulator quantities from (ε_0, n) alone:

Quantity	Formula	Gap	Source
D_0	$n(1 - e^{-1/n})$	0.03%	K2
K^*	D_0/n	0.43%	K1
H_a, H_b	$\sqrt{d_s} \pm n\varepsilon_0$	0.011%	K3
φ_{var}	$1/2 - 1/n^2$	2.3%	Kuramoto
p	$(n+1)/(n+2)$	exact	New
σ_Ψ	$\varepsilon_0(n+2)/\sqrt{2n+3}$	0%	AR(1)+p
ψ_{grad}	$\sigma_\Psi / \ln \varphi$	0.016%	K8
F	$(1 + \varphi_{\text{var}})(1 + \psi_{\text{grad}})$	0.71%	K8 path
κ	$\sqrt{d_s} \ln 2/n$	0.047%	K7
β_{sim} with $m = \varphi$	All above	0.26%	K7+K8+K3+K6

7 Complete Status Table

Result	Status	Note
$D_0 = n(1 - e^{-1/n})$ (K2)	Derived	0.03%
$K^* = D_0/n$ (K1)	Derived	0.43%
$(H_a + H_b)/2 = \sqrt{d_s}$ (K3)	Derived	0.011%
$\varphi_{\text{var}} = 1/2 - 1/n^2$	Derived	2.3%
$\text{Cov}(\psi_{\text{grad}}, \Delta H) = 0$	Derived	Exact via K3
$T_{\text{avg}} = 2nK^*/((n-1)\varepsilon_0)$	Derived	1.2%
$\kappa = \sqrt{d_s} \ln 2/n$ (K7)	Hypothesis	0.047%, new
$p = (n+1)/(n+2)$ (AR(1))	Hypothesis	Exact, needs derivation
$\psi_{\text{grad}} = \sigma_\Psi / \ln \varphi$ (K8)	Hypothesis	0.016%, new
$F \cdot H_u = \sqrt{2}\beta$ (K9)	Hypothesis	0.005%, new
$D_0 m / \varphi_{\text{var}} = \pi$ (K10)	Hypothesis	0.013%, new
$m_{\text{fire}} = \varphi$ (K6)	Hypothesis	0.06–0.63%
$F = D_0 m$	Hypothesis	0.36%
$\beta = d_s / (2 \ln 2)$ algebraic	Open	2.81% with K7+K8
$m(m-1) = 1$ from δ -chain	Open	Energy balance path
p from δ -chain	Open	Group-theory of $n+2$
P0-H Subject 4	Open	Empirical gate

8 Open Questions

Priority 1: AR(1) coefficient $p = (n+1)/(n+2)$ from δ -chain

The Psi update persistence $p = (n+1)/(n+2)$ is exact for $n = 8$ but not yet derived from the δ -chain. The quantity $n+2$ could arise as: (a) n symmetry modes plus 2 (self + noise reference); (b) a group-theoretic count in $\text{SO}(3)$ with $n = 8$ generators; (c) the relaxation time of the Psi field in units of T_{avg} . Deriving p would make σ_Ψ and ψ_{grad} fully Derived.

Priority 2: K7 $\kappa = \sqrt{d_s} \ln 2/n$ physical proof

The argument “one bit per $\sqrt{d_s}$ distributed over n modes” is physically motivated but not yet derived from the δ -chain. The proof would show that the Born resampling step transfers exactly $\alpha\sqrt{d_s}/n$ units of mass from IDV to Ω per R-event. This connects to the IDV formula and the dimensional structure of the resolution scale.

Priority 3: $m(m-1) = 1$ without $s_{\max}$

The self-consistency map gives $m^* \approx \varphi$ at $s_{\max} = 400$ (gap -0.038% with K7). But m^* depends on $s_{\max}$. The energy-balance path: find $f(\varepsilon_0, n, D_0)$ such that $m(m-1) = \beta\kappa/f$ has $m = \varphi$ as its unique solution.

9 Key Numbers — Updated

$\alpha = \ln 2$	$\varepsilon_0 = 0.00755$	$D_0 = 0.9400$	$\kappa_{K7} = 0.15007$
$K^* = 0.11750$	$n = 8$	$N = 500$	$d_s = 3.031 \pm 0.076$
$\beta = 2.212 \pm 0.015$	$F = 1.527$	$T_{\text{avg}} = 35.4$	$\varphi = 1.61803$
$H_a = 1.7925$	$H_b = 1.6717$	$H_u = 2.055$	$f_{II} = 0.147$
$p_{\text{AR1}} = 9/10$	$\sigma_{\Psi} = 0.01732$	$\psi_{\text{grad}} = 0.03600$	$\varphi_{\text{var}} = 0.4844$

$\kappa = \sqrt{3} \ln 2/8$. The crystallisation rate was never free. It was one bit of distinction, spread over three dimensions and eight modes, waiting to be read from δ . K7 reads it.